

Electronic Circuits

(Textbook)

Robert L. Boylestad and Louis Nashelsky

Electronic Devices and Circuit Theory (11th Ed.) Pearson 2013

CHAPTER 1: Semiconductor Diodes

CHAPTER 2: Diode Applications

CHAPTER 3: Bipolar Junction Transistors

CHAPTER 4: DC Biasing–BJTs

CHAPTER 5: BJT AC Analysis

CHAPTER 6: Field-Effect Transistors

CHAPTER 7: FET Biasing

CHAPTER 8: FET Amplifiers

CHAPTER 9: BJT and JFET Frequency Response

CHAPTER 10: Operational Amplifiers

CHAPTER 11: Op-Amp Applications

CHAPTER 12: Power Amplifiers

CHAPTER 13: Linear-Digital ICs

CHAPTER 14: Feedback and Oscillator Circuits

CHAPTER 15: Power Supplies (Voltage Regulators)

CHAPTER 16: Other Two-Terminal Devices

CHAPTER 17: *pnpn* and Other Devices

※ Circuits

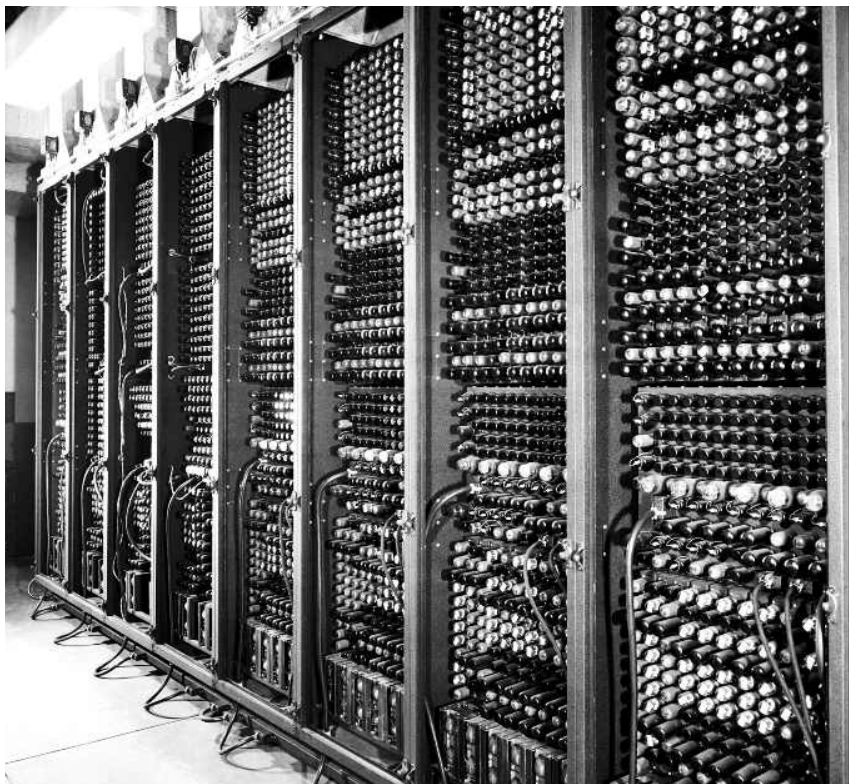
1. Electric Circuits
2. Electronic Circuits
3. Digital Circuits

- Elements of Electronic Circuits

1. Vacuum Tubes

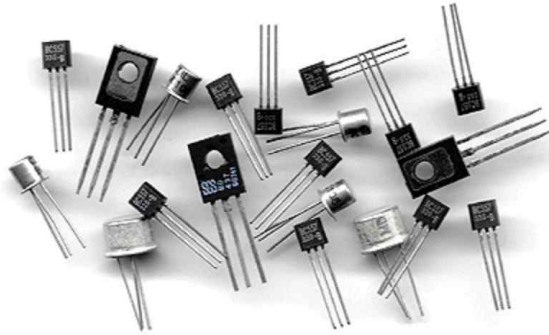


※ 1946 ENIAC Digital Computer using Vacuum Tubes



(24 meters long, 2.6 meters high, 18000 vacuum tubes)

2. Semiconductor Discrete elements - BJT and FET

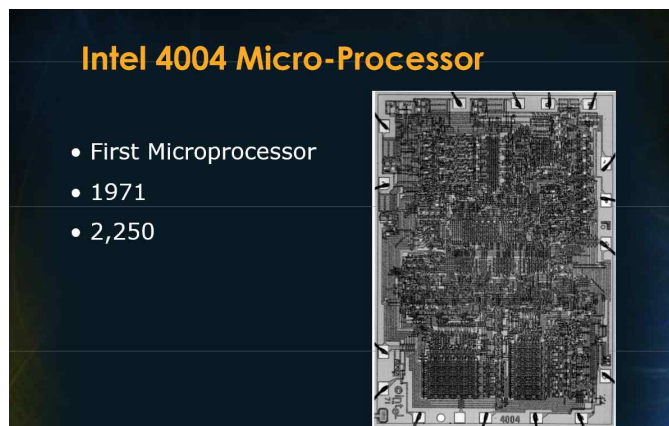


※ Transistor Radio (1960s)

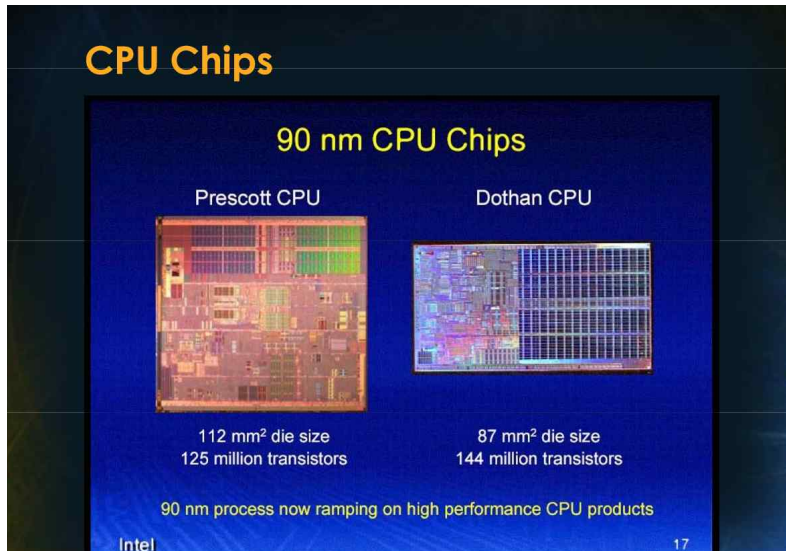


3. Semiconductor Integrated Circuits (IC) - VLSI

- 1971 (Intel) ($10\ \mu\text{m}$ technology)



- June, 2004 (intel)



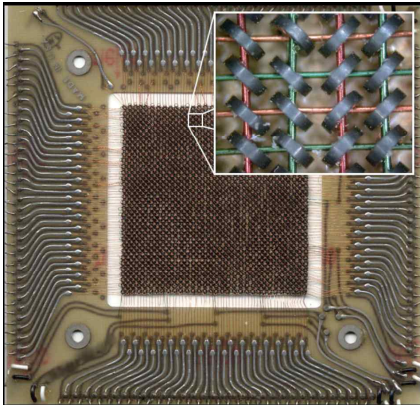
- 2023 Samsung (3 nm technology)

June 21, 2023 Samsung 3 nm technology	※ 1958 IBM Computer
Development of 3nm GAA MBCFET™ Technology More Moore – Gate All Around Planar FinFET MBCFET™ Multi Bridge Channel FET	

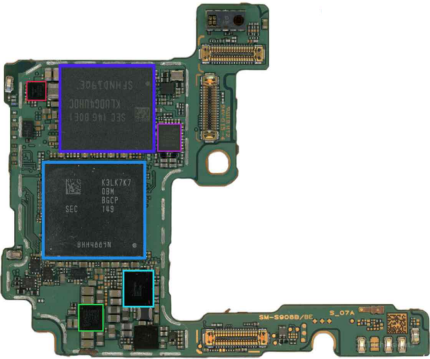
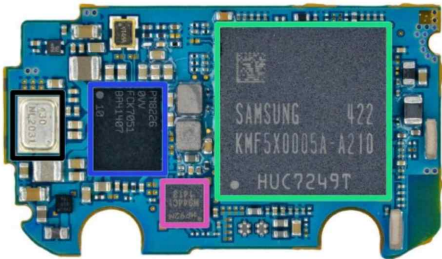
<https://semiconductor.samsung.com/news-events/tech-blog/3nm-gaa-mbcfet-unrivaled-sram-design-flexibility/>

※ <https://semiconductor.samsung.com/>

※ Memory [RAM]

Semiconductor Memory (2 TeraBytes)	※ Magnetic Core Memory 1964 11×11 cm / 64 x 64 bits (4 KBits)
	

※ VLSI (Very Large Scale Integration)

Inside of a smartphone Samsung Galaxy S22 Ultra	Inside of a G4 watch
	

※ Samsung Galaxy S23 Ultra Specifications

Samsung Galaxy S23 Ultra		
	Released 2023, February 17 234g, 8.9mm thickness Android 13, One UI 5.1 256GB/512GB/1TB storage, no card slot	
	68% 7,694,766 HITS 1162 BECOME A FAN	
	6.8" 1440x3088 pixels	
	200MP 4320p	
	8/12GB RAM Snapdragon 8 Gen 2	
	5000mAh Li-Ion	
REVIEW	OPINIONS	COMPARE
PICTURES	PRICES	
NETWORK	Technology	GSM / CDMA / HSPA / EVDO / LTE / 5G
LAUNCH	Announced	2023, February 01
	Status	Available. Released 2023, February 17
BODY	Dimensions	163.4 x 78.1 x 8.9 mm (6.43 x 3.07 x 0.35 in)
	Weight	234 g (8.25 oz)
	Build	Glass front (Gorilla Glass Victus 2), glass back (Gorilla Glass Victus 2), aluminum frame
	SIM	Nano-SIM and eSIM or Dual SIM (2 Nano-SIMs and eSIM, dual stand-by)
		IP68 dust/water resistant (up to 1.5m for 30 min)
		Armor aluminum frame with tougher drop and scratch resistance (advertised)
		Stylus (Bluetooth integration, accelerometer, gyro)
DISPLAY	Type	Dynamic AMOLED 2X, 120Hz, HDR10+, 1200 nits (HBM), 1750 nits (peak)
	Size	6.8 inches, 114.7 cm ² (~89.9% screen-to-body ratio)
	Resolution	1440 x 3088 pixels, 19.3:9 ratio (~500 ppi density)
	Protection	Corning Gorilla Glass Victus 2
		Always-on display
PLATFORM	OS	Android 13, One UI 5.1
	Chipset	Qualcomm SM8550-AC Snapdragon 8 Gen 2 (4 nm)
	CPU	Octa-core (1x3.36 GHz Cortex-X3 & 2x2.8 GHz Cortex-A715 & 2x2.8 GHz Cortex-A710 & 3x2.0 GHz Cortex-A510)
	GPU	Adreno 740
MEMORY	Card slot	No
	Internal	256GB 8GB RAM, 256GB 12GB RAM, 512GB 12GB RAM, 1TB 12GB RAM
		UFS 4.0
MAIN CAMERA	Quad	200 MP, f/1.7, 24mm (wide), 1/1.3", 0.6µm, multi-directional PDAF, Laser AF, OIS
		10 MP, f/4.9, 230mm (periscope telephoto), 1/3.52", 1.12µm, Dual Pixel PDAF, OIS, 10x optical zoom
		10 MP, f/2.4, 70mm (telephoto), 1/3.52", 1.12µm, Dual Pixel PDAF, OIS, 3x optical zoom
		12 MP, f/2.2, 13mm, 120° (ultrawide), 1/2.55", 1.4µm, Dual Pixel PDAF, Super Steady video
	Features	LED flash, auto-HDR, panorama
	Video	8K@24/30fps, 4K@30/60fps, 1080p@30/60/240fps, 1080p@960fps, HDR10+, stereo sound rec., gyro-EIS
SELFIE CAMERA	Single	12 MP, f/2.2, 26mm (wide), Dual Pixel PDAF
	Features	Dual video call, Auto-HDR, HDR10+
	Video	4K@30/60fps, 1080p@30fps
SOUND	Loudspeaker	Yes, with stereo speakers
	3.5mm jack	No
		32-bit/384kHz audio
		Tuned by AKG
COMMS	WLAN	Wi-Fi 802.11 a/b/g/n/ac/6e, tri-band, Wi-Fi Direct
	Bluetooth	5.3, A2DP, LE
	Positioning	GPS, GLONASS, BDS, GALILEO
	NFC	Yes
	Radio	No
	USB	USB Type-C 3.2, OTG
FEATURES	Sensors	Fingerprint (under display, ultrasonic), accelerometer, gyro, proximity, compass, barometer
		Samsung DeX, Samsung Wireless DeX (desktop experience support)
		Bixby natural language commands and dictation
		Samsung Pay (Visa, MasterCard certified)
		Ultra Wideband (UWB) support
BATTERY	Type	Li-Ion 5000 mAh, non-removable
	Charging	45W wired, PD3.0, 65% in 30 min (advertised)
		15W wireless (Qi/PMA)
		4.5W reverse wireless

https://www.gsmarena.com/samsung_galaxy_s23_ultra-12024.php

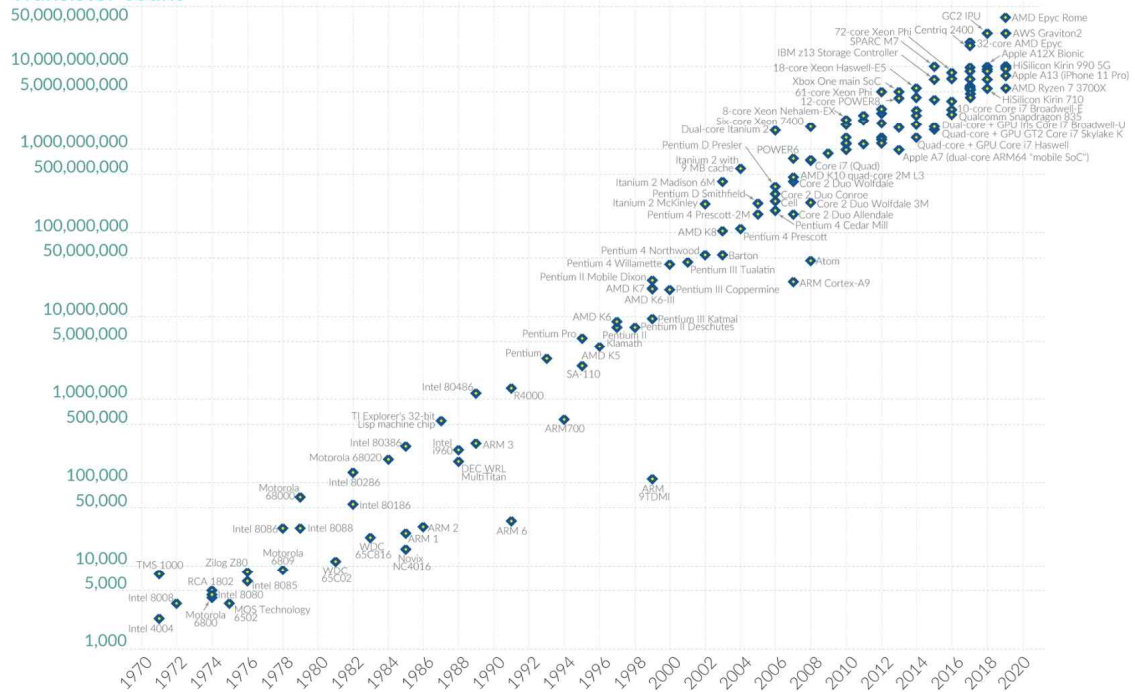
- The number of transistors on a chip

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

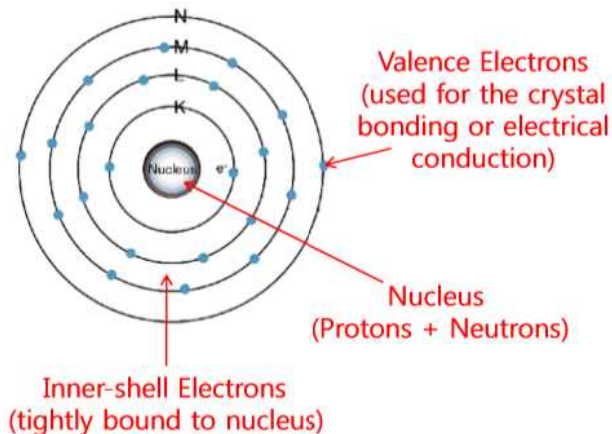
OurWorldinData.org – Research and data to make progress against the world's largest problems.

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counts for microprocessors against dates of introduction, nearly doubling every two years

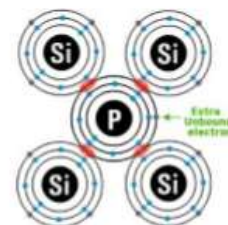
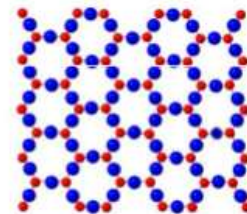
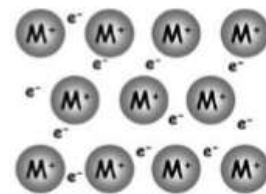
SEMICONDUCTOR & DIODES

● Very Simple Model of Atoms



- Conductor, Semiconductor and Insulator

- Conductor (Ag, Cu, Au, Al, ...)
 - Valence electrons are NOT used in the bond, but they form "**sea of electrons**"
 - Even with small V applied, large I is created due to lots of "moving" charges. Thus, the electrical resistance (R) is low.
- Insulator (SiO_2 , Al_2O_3 , ...)
 - All valence electrons are **tightly bound** for the crystal bonding.
 - Even with large V applied, very small I is created due to very little available "moving" charges. Thus, R is very high.
- Semiconductor (Si, GaN, SiGe, GaAs, ...)
 - Most of valence electrons are used for the crystal bonding. But **some of them are released** from the atom and can freely move within the lattice.



Periodic Table of the Elements

<http://chemistry.about.com>
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About Chemistry

1A 1 H 1.00794 Hydrogen	2A 4 He 4.002602 Helium	3A 5 B 10.811 Boron	4A 6 C 12.0107 Carbon	5A 7 N 14.0067 Nitrogen	6A 8 O 15.9994 Oxygen	7A 9 F 18.9984032 Fluorine	8A 10 Ne 20.1797 Neon
3 Li 6.941 Lithium	12 Mg 24.3050 Magnesium	13 Al 26.9815386 Aluminum	14 Si 28.0855 Silicon	15 P 30.973762 Phosphorus	16 S 32.065 Sulfur	17 Cl 35.453 Chlorine	18 Ar 39.948 Argon
19 K 39.0983 Potassium	20 Ca 40.078 Calcium	21 Sc 44.955912 Scandium	22 Ti 47.887 Titanium	23 V 50.9415 Vanadium	24 Cr 51.9961 Chromium	25 Mn 54.938045 Manganese	26 Fe 55.845 Iron
37 Rb 85.4678 Rubidium	38 Sr 87.62 Strontium	39 Y 88.90585 Yttrium	40 Zr 91.224 Zirconium	41 Nb 92.90638 Niobium	42 Mo 95.96 Molybdenum	43 Tc [98] Technetium	44 Ru 101.07 Ruthenium
55 Cs 132.9054519 Cesium	56 Ba 137.327 Barium	57-71 Lanthanides	72 Hf 178.49 Hafnium	73 Ta 180.94788 Tantalum	74 W 183.84 Tungsten	75 Re 186.207 Rhenium	76 Os 190.23 Osmium
87 Fr [223] Francium	88 Ra [226] Radium	89-103 Actinides	104 Rf [261] Rutherfordium	105 Db [262] Dubnium	106 Sg [266] Seaborgium	107 Bh [264] Bohrium	108 Hs [277] Hassium
111 Uut [284] Ununtrium	112 Uup [285] Ununpentium	113 Uut [284] Ununtrium	114 Uuq [289] Ununquadium	115 Uup [288] Ununpentium	116 Uuo [294] Ununseptium	117 Uus [294] Ununseptium	118 Uuo [294] Ununseptium
119 Uue [285] Ununennium	120 Uub [286] Unbium	121 Uut [289] Ununtrium	122 Uuq [290] Ununquadium	123 Uup [291] Ununpentium	124 Uuo [292] Ununseptium	125 Uus [293] Ununseptium	126 Uuo [294] Ununseptium

57 La 138.90547 Lanthanum	58 Ce 140.116 Cerium	59 Pr 140.90765 Praseodymium	60 Nd 144.242 Neodymium	61 Pm [145] Promethium	62 Sm 150.36 Samarium	63 Eu 151.964 Europium	64 Gd 157.25 Gadolinium	65 Tb 158.92535 Terbium	66 Dy 162.500 Dysprosium	67 Ho 164.93032 Holmium	68 Er 167.259 Erbium	69 Tm 168.93421 Thulium	70 Yb 173.054 Ytterbium	71 Lu 174.9668 Lutetium
89 Ac [227] Actinium	90 Th 232.03806 Thorium	91 Pa 231.03688 Protactinium	92 U 238.02891 Uranium	93 Np [237] Neptunium	94 Pu [244] Plutonium	95 Am [243] Americium	96 Cm [247] Curium	97 Bk [247] Berkelium	98 Cf [251] Californium	99 Es [252] Einsteinium	100 Fm [257] Fermium	101 Md [258] Mendelevium	102 No [259] Nobelium	103 Lr [262] Lawrencium

Alkali Metals	Alkaline Earth	Transition Metal	Basic Metal	Semi Metal	Non Metal	Halogen	Noble Gas	Lanthanides	Actinides
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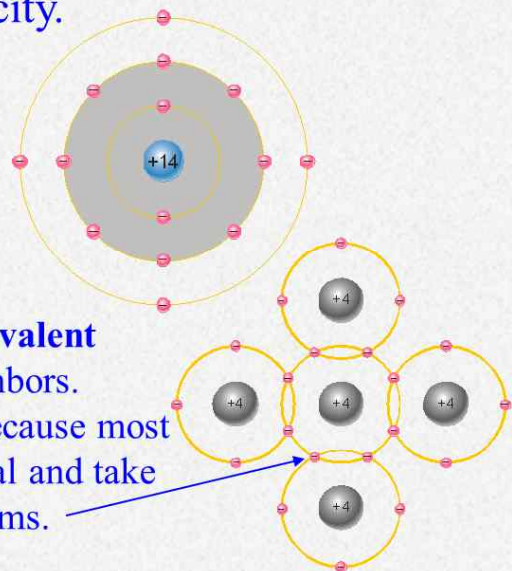
● Semiconductors

Semiconductors are between conductors and insulators in their ability to conduct electricity.

Silicon is an example of a single element **semiconductor**. It has four electrons in its valence band.

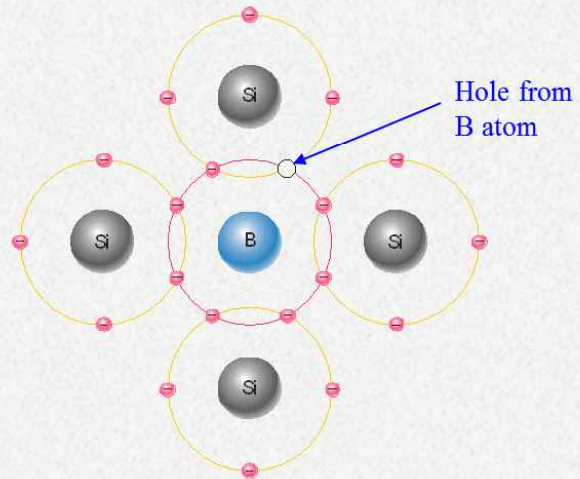
Unlike metals, silicon forms strong **covalent** bonds (shared electrons) with its neighbors.

Intrinsic silicon is a poor conductor because most of the electrons are bound in the crystal and take part in forming the bonds between atoms.



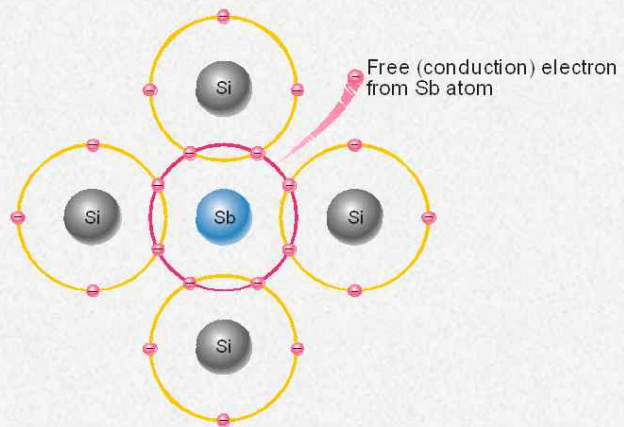
- Impurity and p-type material

An impurity such as boron leaves a vacancy in the valence band, creating a *p*-material. Both *p*- and *n*-materials have energy levels that are different than intrinsic silicon.



- Impurity and n-type material

Certain impurities will change the conductivity of silicon. An impurity such as Antimony has an electron that is not part of the bonding electrons so is free. This creates an *n*-material.



- Semiconductor Materials

Materials commonly used in semiconductor devices

1. Silicon (Si)
2. Germanium (Ge)
3. Gallium Arsenide (GaAs)

- Doping

The electrical characteristics of silicon and germanium are improved by adding materials in a process called *doping*.

There are just two types of doped semiconductor materials:

***n*-type**

***n*-type** materials contain an excess of conduction band electrons.

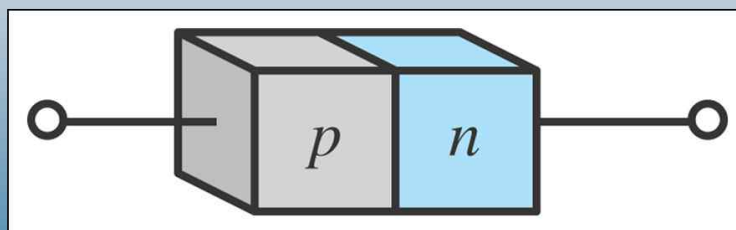
***p*-type**

***p*-type** materials contain an excess of valence band holes.

- p-n junctions and Semiconductor Diode

***p-n* Junctions**

One end of a silicon or germanium crystal can be doped as a *p*-type material and the other end as an *n*-type material.



The result is a ***p-n* junction**

n-type material : excessive electrons

p-type material : deficient electrons (i.e., excessive holes)

- Diode Operating Conditions

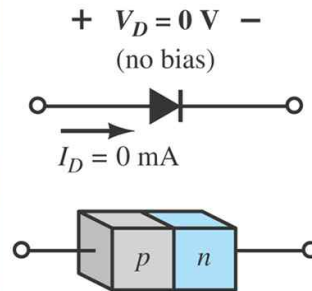
No Bias, Reverse Bias, Forward Bias

(1) No Bias

No external voltage is applied: $V_D = 0 \text{ V}$

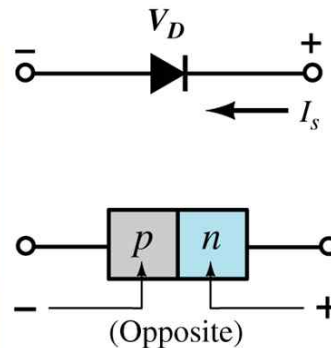
There is no diode current: $I_D = 0 \text{ A}$

Only a modest depletion region exists



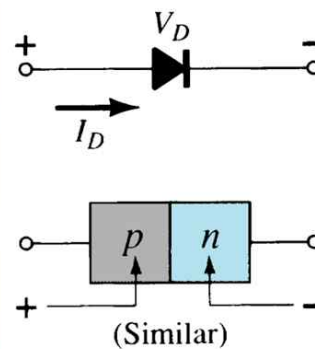
(2) Reverse Bias

External voltage is applied across the p - n junction in the opposite polarity of the p - and n -type materials.

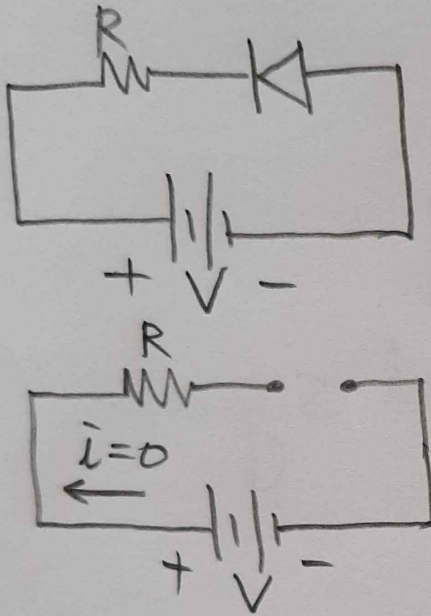


(3) Forward Bias

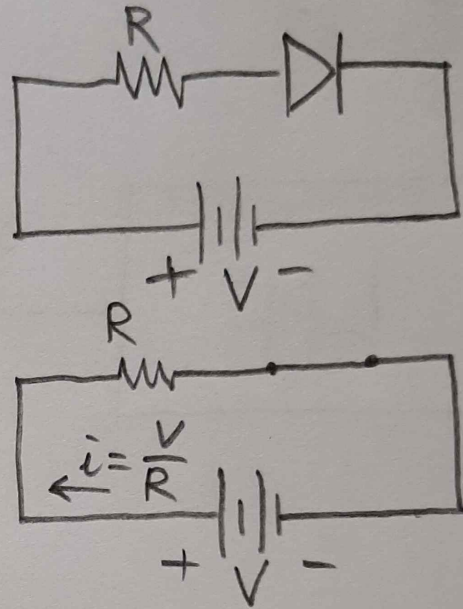
External voltage is applied across the p - n junction in the same polarity as the p - and n -type materials.



reverse bias



forward bias



- Forward-Bias and Reverse-Bias of Semiconductor Diodes



- Increase contact potential
- Cause the drift of "minority" carriers through the depletion region.

➡ Reverse current:

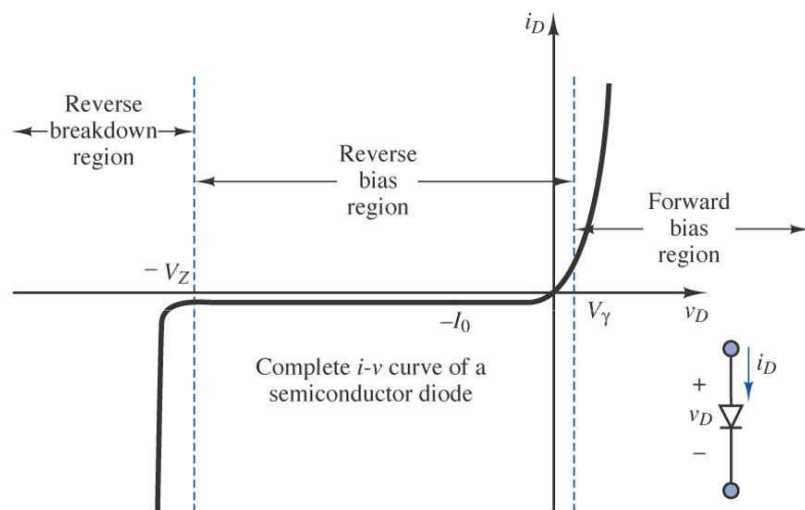
- Small
- Independent of voltage applied
- determined by thermal carrier generation

- Lower contact potential
- Help the diffusion of "majority" carriers through the depletion region.

➡ Forward current:

- Large
- Dependent on the voltage applied

- I-V Characteristics of Semiconductor Diode



Reverse breakdown:

- Avalanche breakdown
- Impact Ionization
- Zener breakdown

Reverse current:

- Small
- Independent of voltage applied

Forward current:

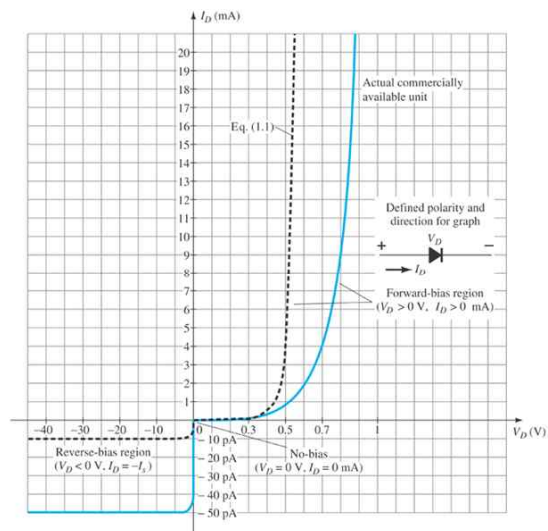
- Large
- Dependent on the voltage applied

※ Peak Inverse Voltage (PIV)

At some point the reverse bias voltage is so large, the diode breaks down and the reverse current increases dramatically.

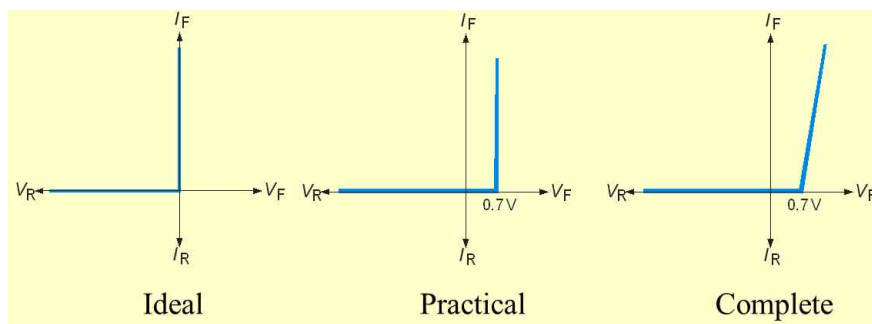
- Actual Diode Characteristics

※ Note the regions for three bias conditions



(Carefully note the scale for each of these conditions)

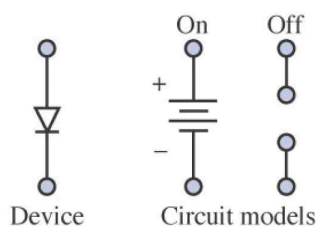
- Three Diode Approximations



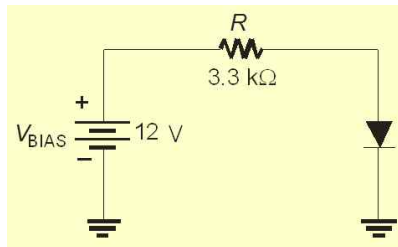
※Forward Bias Voltage (Threshold Voltage)

Si Diode	0.7 V (or 0.6 V)
Ge Diode	0.2 V
GaAs Diode	1.2 V

- Practical Model of Diode



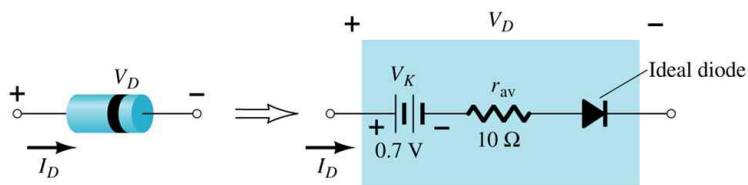
(Example) Use the practical model to determine the current.



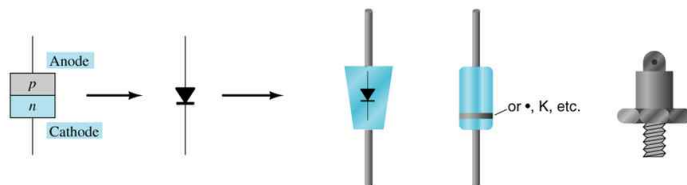
$$V_R = V_{\text{BIAS}} - 0.7 \text{ V} = 12 \text{ V} - 0.7 \text{ V} = 11.3 \text{ V}$$

$$I = \frac{V_R}{R} = \frac{11.3 \text{ V}}{3.3 \text{ k}\Omega} = 3.4 \text{ mA}$$

- Diode Equivalent Circuit

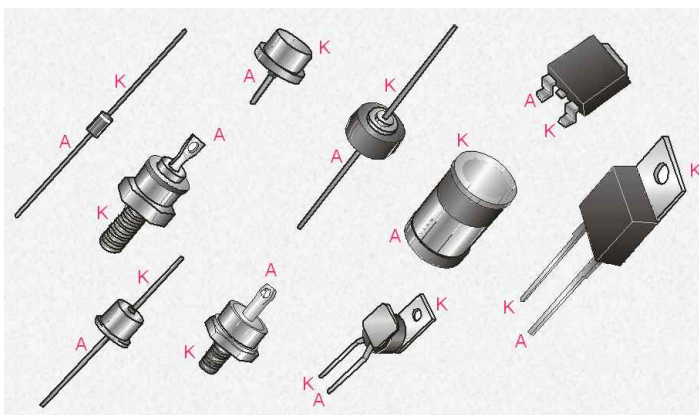


- Diode Symbol and packaging



※ A : anode, K : cathode

- Typical Diode Packages



- Zener Diode

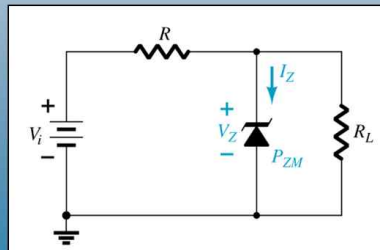
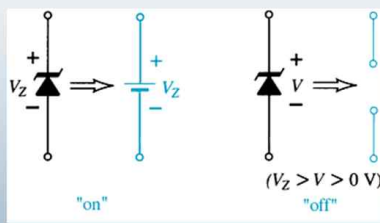
The Zener is a diode that is operated in reverse bias at the Zener Voltage (V_Z).

When $V_i \geq V_Z$

- The Zener is on
- Voltage across the Zener is V_Z
- Zener current: $I_Z = I_R - I_{RL}$
- The Zener Power: $P_Z = V_Z I_Z$

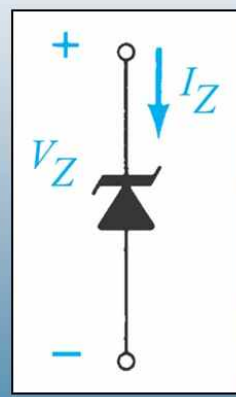
When $V_i < V_Z$

- The Zener is off
- The Zener acts as an open circuit



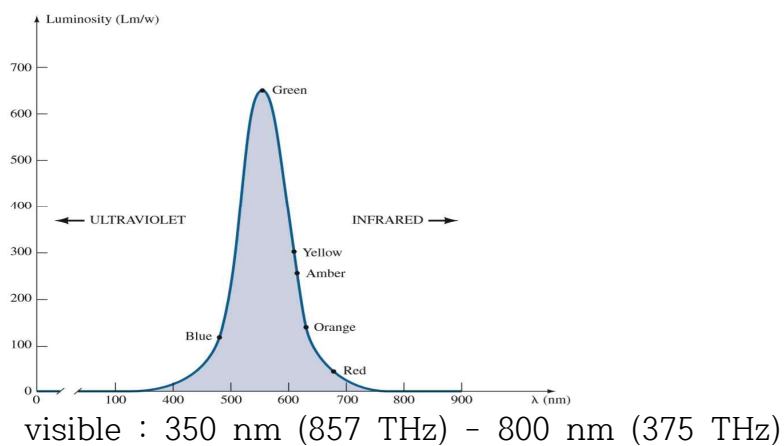
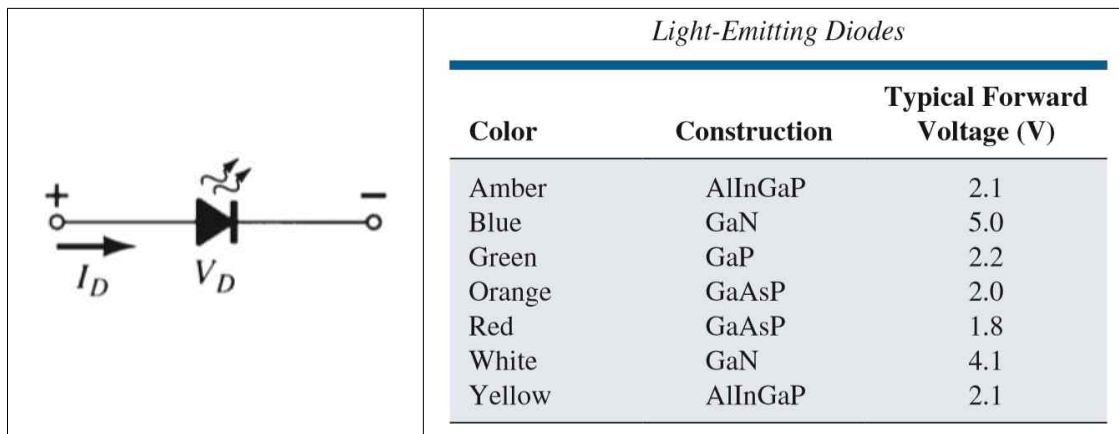
A **Zener diode** is one that is designed to safely operate in its zener region; i.e., biased at the Zener voltage (V_Z).

Common zener diode voltage ratings are between 1.8 V and 200 V

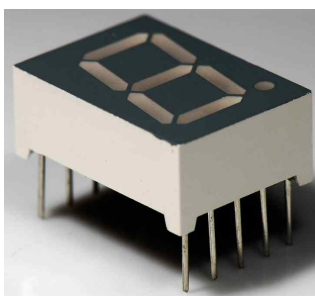


※ Zener diodes are used for the Constant Voltage Power Supply.

- Light Emitting Diode (LED)



- Applications of LEDs



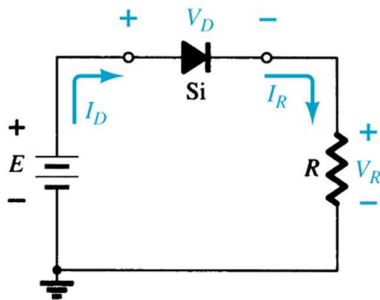
※ LCD : Liquid Crystal Display

Diode Applications

※ We will use silicon diodes. ($V_D = 0.7\text{V}$)

● Rectifier Circuits

(i) Forward Bias



$$V_D = 0.7\text{V}$$

$$V_R = E - V_D$$

$$I_D = I_R = \frac{V_R}{R}$$

(ii) Reverse Bias

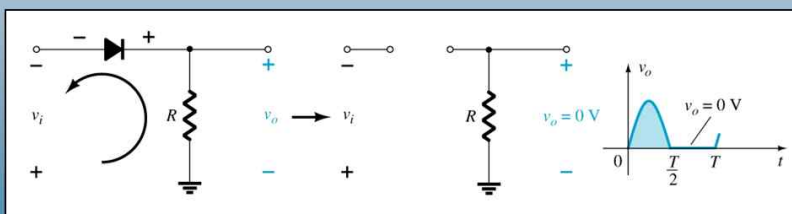
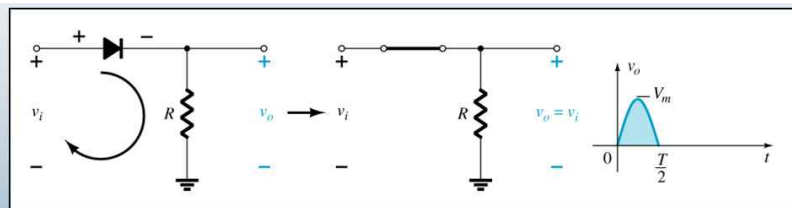
Diodes ideally behave as open circuits.

$$\therefore I_D = I_R = 0$$

- Half-wave Rectification

※ Rectification : AC $\rightarrow$ DC

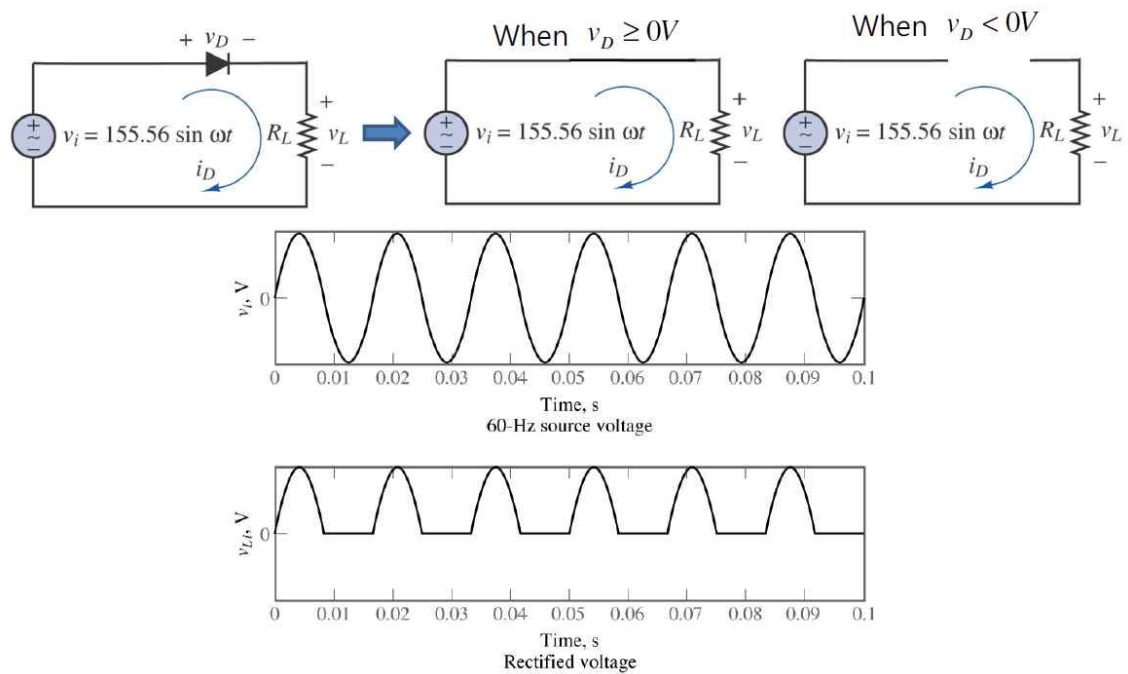
The diode conducts only when it is forward biased, therefore only half of the AC cycle passes through the diode to the output.



The DC output voltage is $0.318V_m$, where V_m = the peak AC voltage.

$$\text{※ DC output voltage} = \frac{1}{T} \int_0^{\frac{T}{2}} V_m \sin \omega t \, dt = 0.318 V_m$$

※ Half-wave Rectifier Circuit by an Ideal Diode

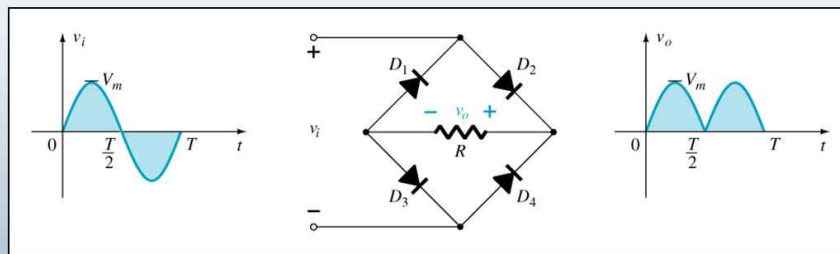


※ DC output voltage (average voltage)

$$\begin{aligned}
 V_{dc} &= \frac{1}{T} \int_0^{\frac{T}{2}} V_m \sin \omega t \, dt = 0.318 V_m \\
 &= 0.318 \times 155.56 \\
 &= 49.47 \text{ [V]}
 \end{aligned}$$

- Full-wave Rectification

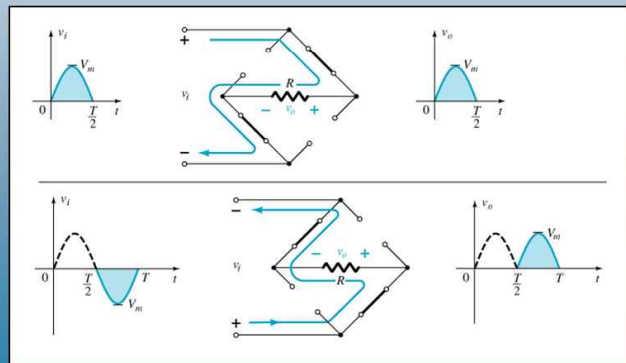
(1) Bridge Rectifier



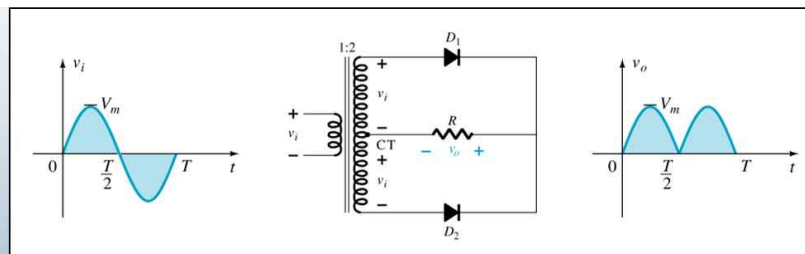
Bridge Rectifier

A full-wave rectifier with four diodes that are connected in a bridge configuration

$$V_{DC} = 0.636 V_m$$



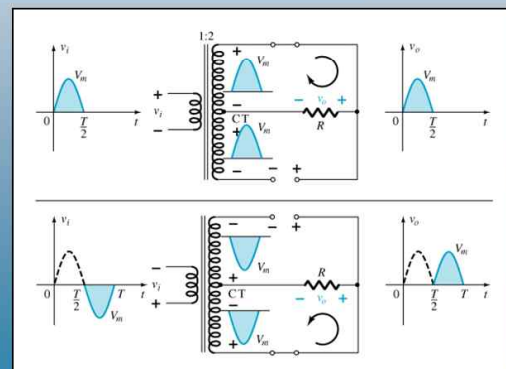
(2) Center-tapped Transformer Rectifier



Center-Tapped Transformer Rectifier

Requires two diodes and a center-tapped transformer

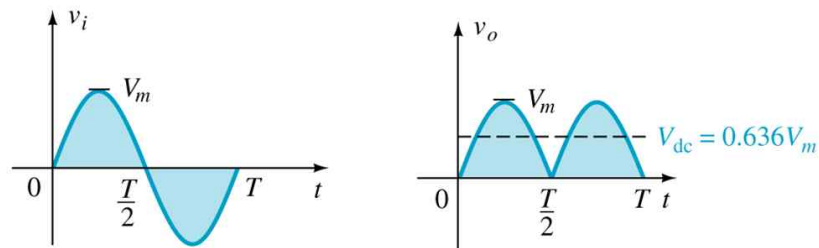
$$V_{DC} = 0.636 V_m$$



- Average values of outputs

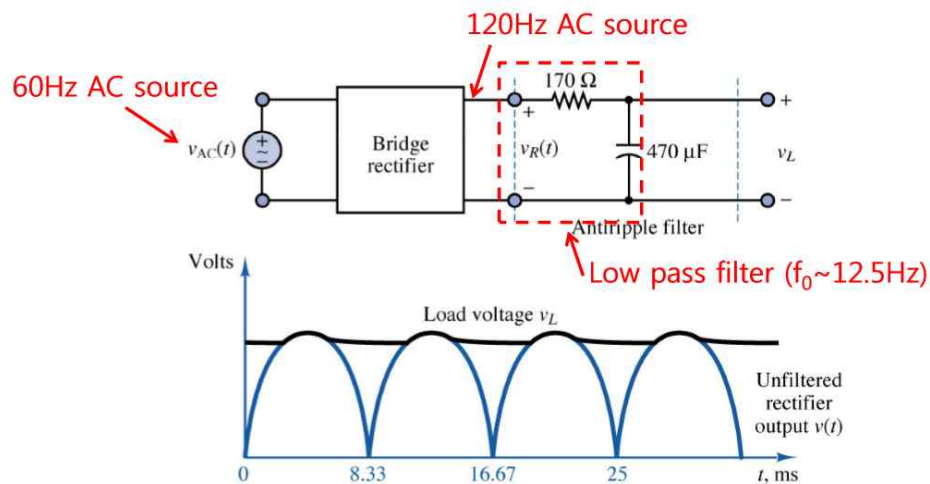
Half-wave rectifier : $0.318 V_m$

Full-wave rectifier : $0.636 V_m$



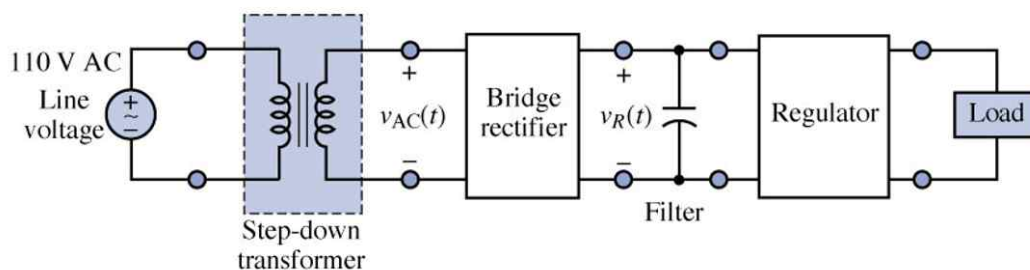
- Reduction of Ripples

- Rectifier circuit controls the polarity of output signal, but the fluctuation of output signal still exists. This is called "ripple".
- This is not desirable and makes the rectifier far from the true DC supply.
- Low-pass filter at the output of rectifier helps the reduction of ripple.



- DC power supply

- For constant output voltage, the DC power supplies should have transformer, bridge rectifier, low-pass filter, and regulator.



※ Smartphone Battery Charger

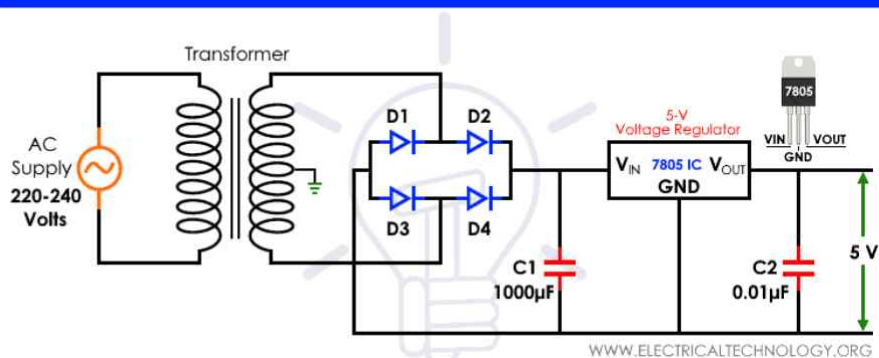
Rectifier : AC $\rightarrow$ DC

input : AC 100-240V

output : DC 5.0V or DC 9.0V

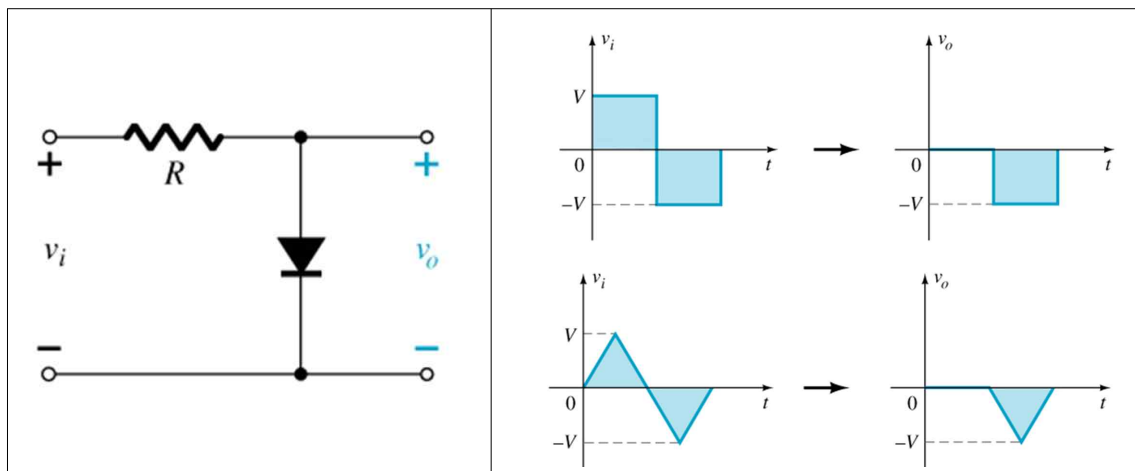
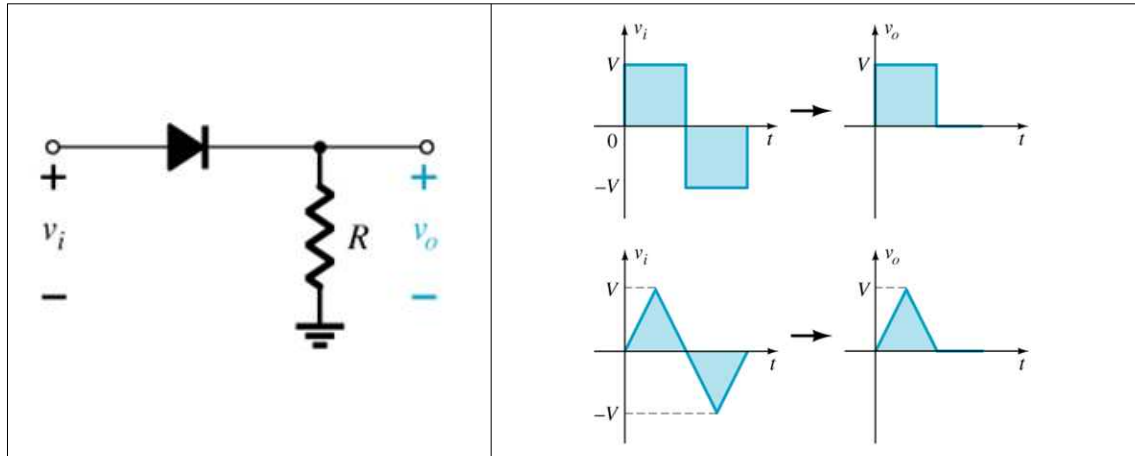


5V Cell Phone Charger Circuit - 230V

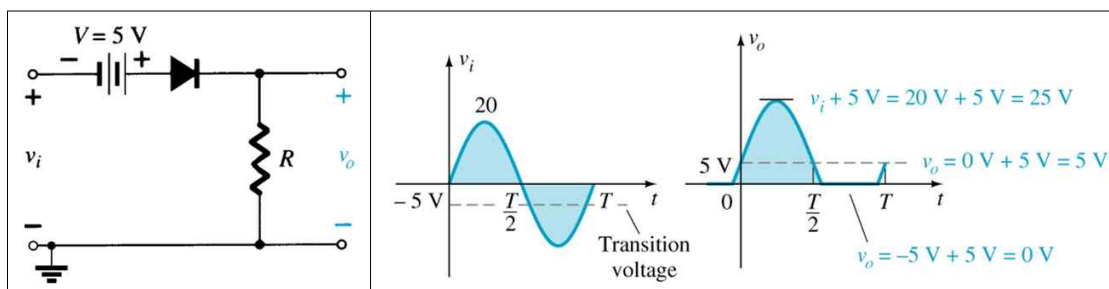


● Clipper Circuits

- Diode Clippers

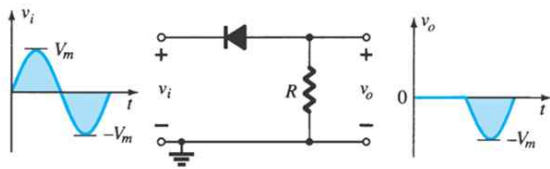


- Biased Clippers

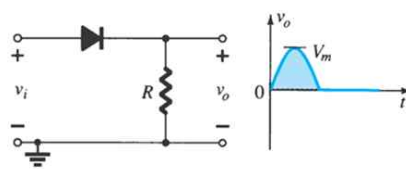


Simple Series Clippers (Ideal Diodes)

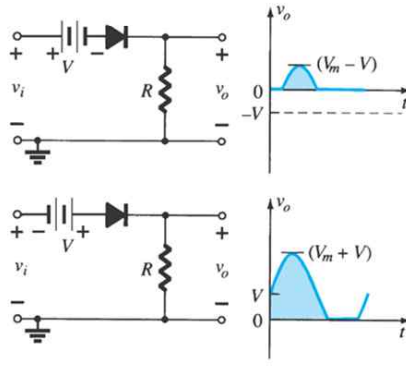
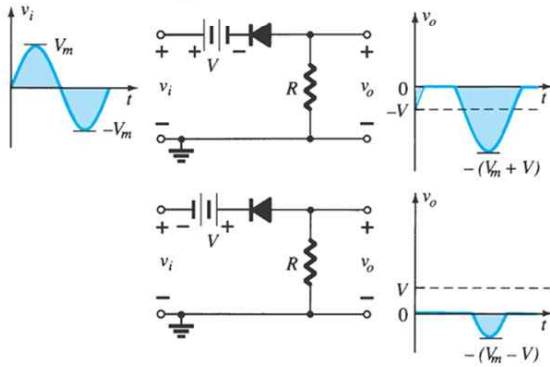
POSITIVE



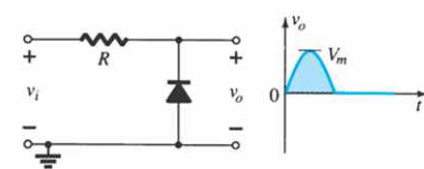
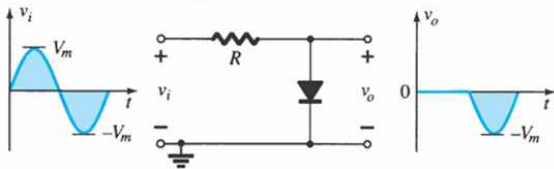
NEGATIVE



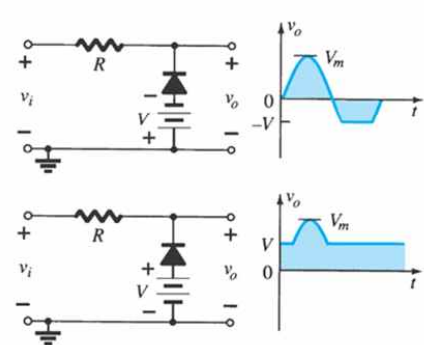
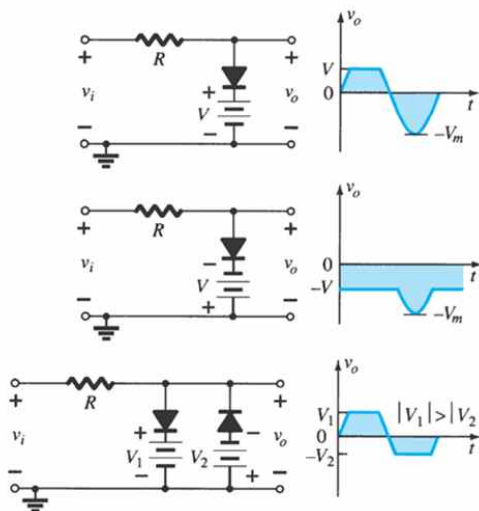
Biased Series Clippers (Ideal Diodes)



Simple Parallel Clippers (Ideal Diodes)

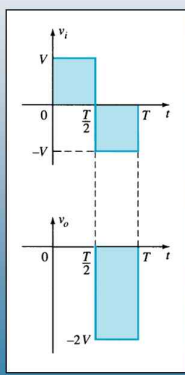
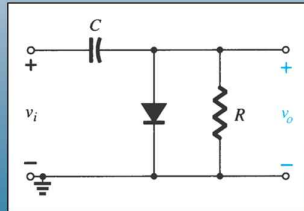


Biased Parallel Clippers (Ideal Diodes)



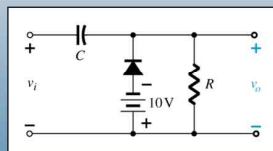
● Clamper Circuits

A diode and capacitor can be combined to “clamp” an AC signal to a specific DC level.

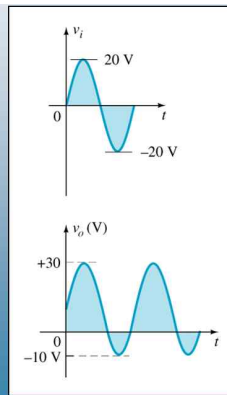


– Basic Clamper Circuits

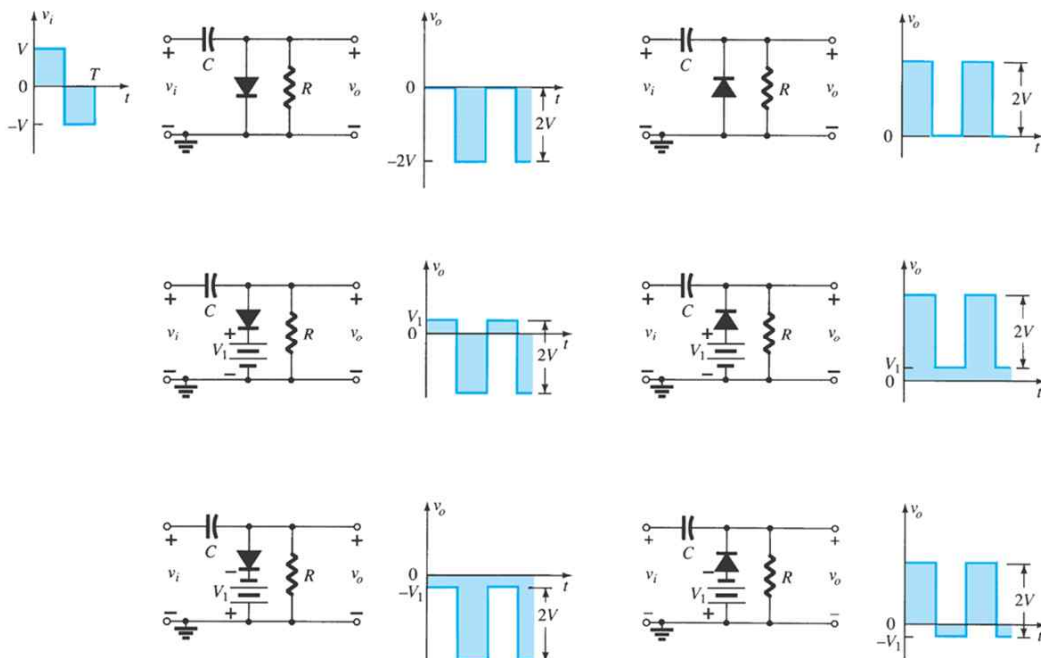
The input signal can be any type of waveform such as a sine, square, or triangle wave.



The DC source lets you adjust the DC clamping level.

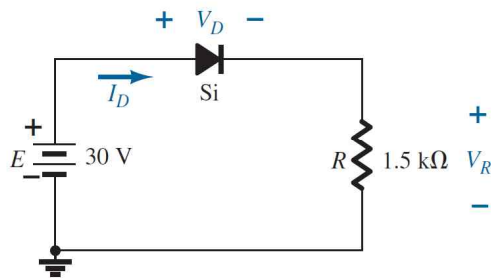


– Summary of Clamper Circuits



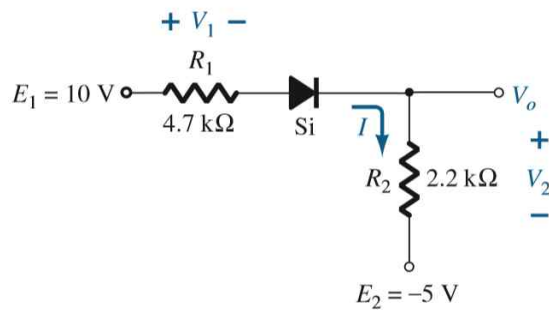
[Examples]

[1] Determine V_D , V_R and I_D .



[2] Example 2.9

Determine I , V_1 , V_2 and V_o .



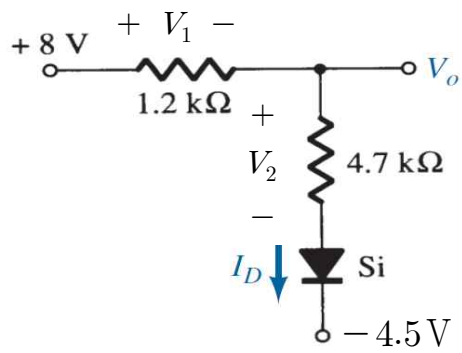
(solution)

$$I = \frac{E_1 - E_2 - V_D}{R_1 + R_2} = 2.07 \text{ mA}$$

$$V_1 = 9.73 \text{ V} , \quad V_2 = 4.55 \text{ V} ,$$

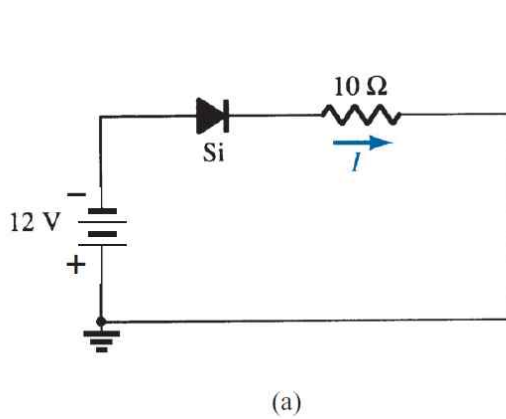
$$V_2 = V_o - E_2 \quad \therefore V_o = V_2 + E_2 = -0.45 \text{ V}$$

[3] Determine V_1 , V_2 , V_o and I_D .

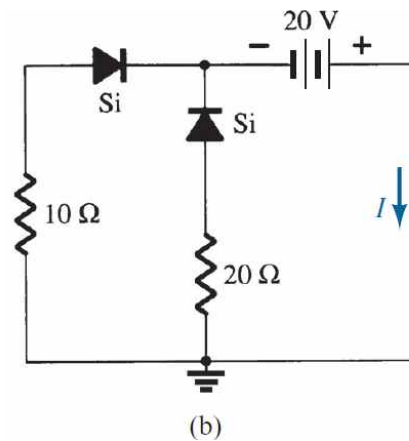


[4] Chap.2 Problem 5 (p.123)

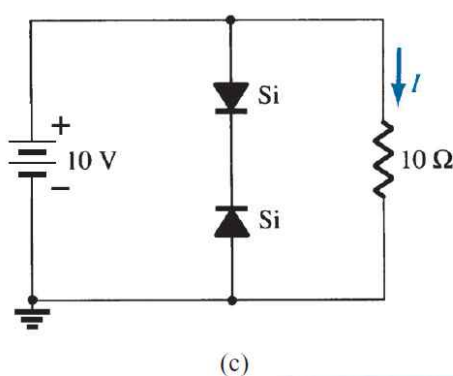
5. Determine the current I for each of the configurations of Fig. 155 using the approximate equivalent model for the diode.



(a) $I = 0 \text{ mA}$; diode reverse-biased.



$$\begin{aligned}
 (b) \quad V_{20\Omega} &= 20 \text{ V} - 0.7 \text{ V} = 19.3 \text{ V} \\
 I(20\Omega) &= \frac{19.3 \text{ V}}{20\Omega} = 0.965 \text{ A} \\
 V(10\Omega) &= 20 \text{ V} - 0.7 \text{ V} = 19.3 \text{ V} \\
 I(10\Omega) &= \frac{19.3 \text{ V}}{10\Omega} = 1.93 \text{ A} \\
 I &= I(10\Omega) + I(20\Omega) \\
 &= 2.895 \text{ A}
 \end{aligned}$$



$$(c) \quad I = \frac{10 \text{ V}}{10\Omega} = 1 \text{ A}; \text{ center branch open}$$

[5] (Example 2.13)

Determine I_1 , I_2 and I_{D_2} .

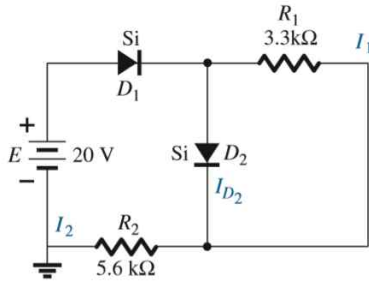


FIG. 37

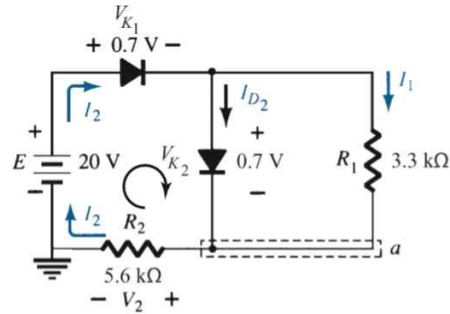


FIG. 38

Solution: The applied voltage (pressure) is such as to turn both diodes on, as indicated by the resulting current directions in the network of Fig. 38. Note the use of the abbreviated notation for “on” diodes and that the solution is obtained through an application of techniques applied to dc series-parallel networks. We have

$$I_1 = \frac{V_{K_2}}{R_1} = \frac{0.7 \text{ V}}{3.3 \text{ k}\Omega} = \mathbf{0.212 \text{ mA}}$$

Applying Kirchhoff's voltage law around the indicated loop in the clockwise direction yields

$$-V_2 + E - V_{K_1} - V_{K_2} = 0$$

and
$$V_2 = E - V_{K_1} - V_{K_2} = 20 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = \mathbf{18.6 \text{ V}}$$

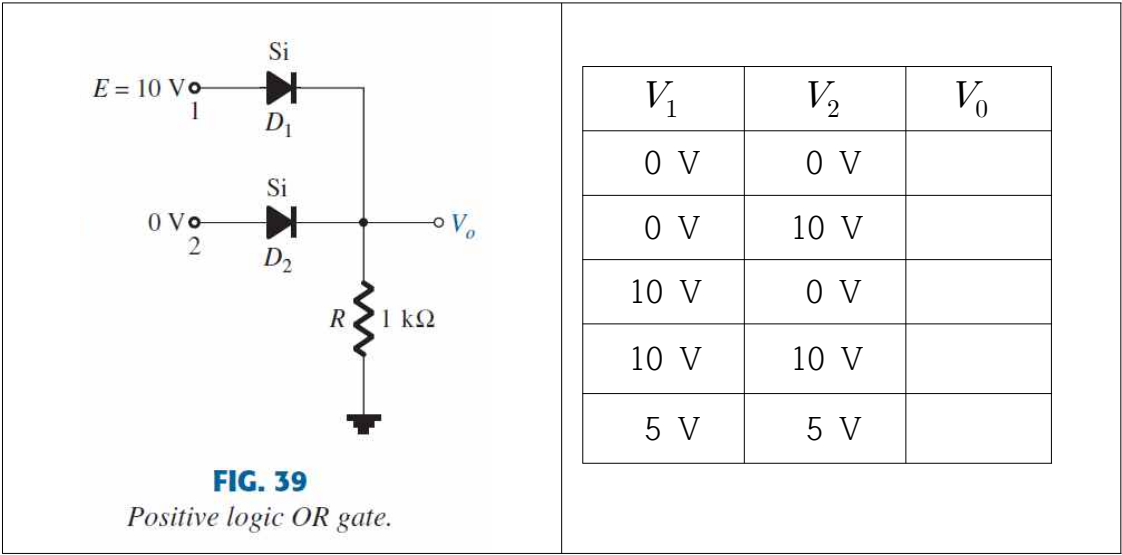
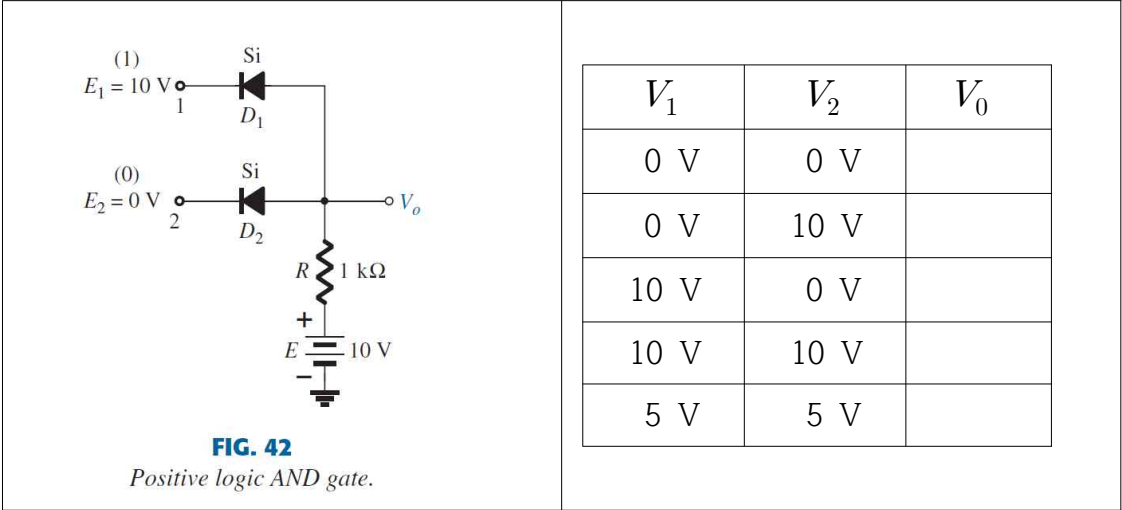
with
$$I_2 = \frac{V_2}{R_2} = \frac{18.6 \text{ V}}{5.6 \text{ k}\Omega} = \mathbf{3.32 \text{ mA}}$$

At the bottom node a ,

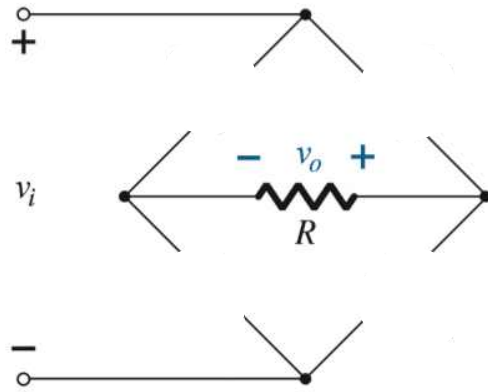
$$I_{D_2} + I_1 = I_2$$

and
$$I_{D_2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} \cong \mathbf{3.11 \text{ mA}}$$

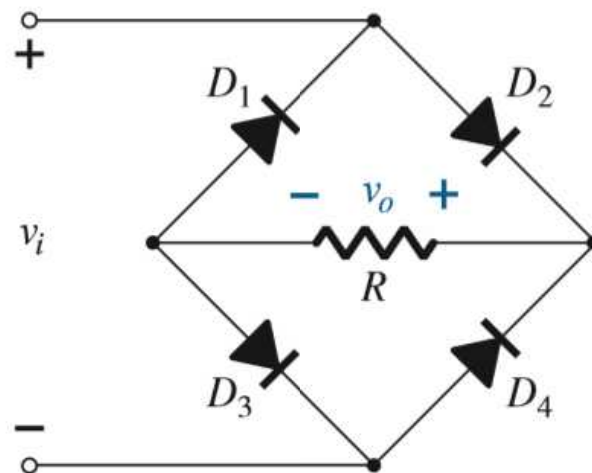
[6] Chap.2 Problem 16-17 (p.125)



[7] Please make a bridge-type full-wave rectifier using 4 diodes.

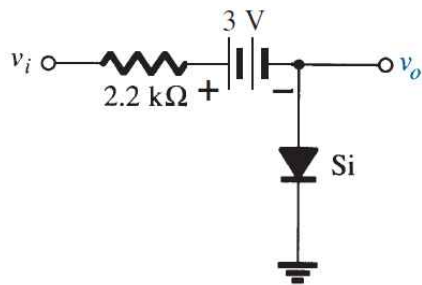
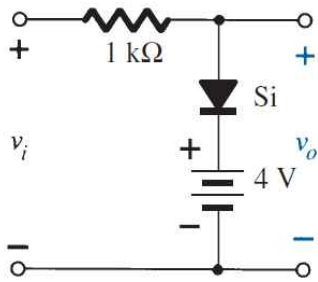


(solution)



[8] Chap.2 Problem 35 (p.128)

Determine v_o , when $v_i = 10\text{V}$, 5V , 0V , -5V and -10V .



[9] Chap.2 Problem 34, 36 (p.128)

Determine v_o , when $v_i = 10\text{V}$, 5V , 0V , -5V and -10V .

